Data Analytics (DA)

Module -1 - Excel

Excel Assignment 1 – Data Exploration

Assignment Task Overview

Problem Statement

You are a **Data Analyst,** and you are required to perform **basic data exploration and analysis in Microsoft Excel**. The dataset contains information about different products, including their **Product IDs, Product Name, Brand Name, Quantity, Categories, and Prices**.

In this assignment, you will use **Excel formulas and functions** to perform calculations, categorize data, and extract useful information from text fields. The goal is to build foundational skills in **data exploration, logical functions, conditional aggregation, and text manipulation** using Excel.

The focus of this assignment is on **data summarization, conditional logic, and text extraction techniques**, which are essential for preparing data for further analysis.

Dataset Description

Dataset: [Product Dataset]

Attributes: *Product ID, Product Name, Brand Name, Quantity, Category, Price*

Tasks:

Basic Data Exploration

1. **Sum, Count, and Average**

   Perform the following calculations using Excel functions:

   - Calculate the **total price of all products** in the dataset.
   - Determine **how many products exist in the dataset.**
   - Calculate the **average price of the products**.

2. **Minimum and Maximum Values**

   Using Excel functions:

   - Identify the **minimum price** among all products.
   - Identify the **maximum price** among all products.

3. **Logical Function – IF**
    - Create a new column named **Price Range**.
    - Use the **IF function** to categorize products based on their price:
        - If **Price ≥ $500** → classify as **High Price**
        - If **Price < $500** → classify as **Standard Price**
4. **Conditional Functions – SUMIF and COUNTIF**

    Perform the following conditional calculations:

    - Use the **SUMIF function** to calculate the **total price of products in the Electronics category**.
    - Use the **COUNTIF function** to determine the **number of products with a price less than $100**.
5. **Text Formatting – LEFT, RIGHT, MID**
    - Use text functions to extract information from the **Product ID** column.
    - Create the following new columns:
        - Create a new column named **Day** with the first 2 characters of each **'Product ID**' using the **LEFT function**.
        - Create a new column named **Country Code** by extracting the last 2 characters from the **'Product ID'** column using the **RIGHT function**.
        - Create a new column named **Month** by extracting the 4th to 6th characters from the **'Product ID'** column using the **MID function**.

Deliverables:-

Submit a project folder containing:

1. **Excel File (.xlsx)**

   The Excel workbook should include:

   - Completed calculations using formulas
   - Newly created columns
   - Correct application of Excel functions
   - Clearly labeled columns and results

2. **PDF Document**

   Include screenshots of:

   - Calculated results for each function
   - Newly created columns
   - Applied formulas in the formula bar

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
Data Summarization	Data Analysis	SUM, COUNT, AVERAGE	Beginner
Data Range Analysis	Statistical Functions	MIN, MAX	Beginner
Logical Functions	Business Logic	IF Conditions	Beginner
Conditional Aggregation	Data Filtering	SUMIF, COUNTIF	Beginner
Text Processing	Data Transformation	LEFT, RIGHT, MID	Beginner
Spreadsheet Skills	Excel Proficiency	Formula Application	Beginner

Rubric & Mark Allocation (Total 10 Marks)

Skill / Component	Marks	Criteria for Full Marks	Partial Credit Description	Partial Marks Range
Basic Calculations	2	Correct SUM, COUNT, AVERAGE results	Minor formula errors	1
Min & Max Analysis	1	Correct use of MIN and MAX	Incorrect cell references, Minor formula errors	0.5
Logical Function (IF)	3	Correct classification of price range	Logical condition errors	1-2
Conditional Functions	3	Correct SUMIF and COUNTIF formulas	Partial calculation errors	1-2
Text Functions	1	Correct extraction using LEFT, RIGHT, and MID	Minor syntax errors	0.5

DA Skill Taxonomy Structure

- **Data Exploration:** Summarizing and analyzing numerical data
- **Logical Analysis:** Applying IF conditions to categorize data
- **Conditional Aggregation:** Using SUMIF and COUNTIF functions
- **Text Manipulation:** Extracting structured information from text fields
- **Spreadsheet Proficiency:** Building structured analytical worksheets

Evaluation Guidelines

- Verify formulas used for each calculation.
- Check that cell references are correctly applied.
- Ensure newly created columns contain accurate results.
- Confirm that Excel functions are used appropriately.
- Assign marks based on accuracy, correctness, and clarity of spreadsheet structure.

Achievement Criteria & Passing Rules

- Students must obtain **at least partial marks in every section**.
- Correct application of **Excel formulas and functions** is mandatory.
- The spreadsheet must be **well-organized and clearly labeled**.
- Documentation screenshots must clearly show formulas and results.